Joseph Bui

Senior Software Engineer

San Francisco, CA
+1 (224) 300-4521
joseph.bui1110@outlook.com
https://www.linkedin.com/in/jnvbui

Results-driven Senior Data Engineer with 9+ years of experience delivering scalable **data engineering solutions** across healthcare and high-growth platforms, including real-time analytics and large-scale data pipelines for ride optimization and social graph insights projects. Expert in **Azure data ecosystem**, **data pipelines**, and **dimensional modeling**, consistently improving data reliability and business decision-making through robust architectures. Proven ability to design **lakehouse architectures**, integrate complex API data sources, and optimize SQL-driven transformations for high-volume systems. Strong track record mentoring engineers and driving adoption of modern **data strategies** aligned with business outcomes.

CORE SKILLS

- **Data Engineering & Azure** : Azure Data Factory, Azure Data Lake Storage ADLS, Azure Databricks, Azure Synapse Analytics, Azure SQL Database, Microsoft Fabric, Spark, Lakehouse Architecture
- **Data Modeling & SQL** : Advanced SQL, Dimensional Modeling, Star Schema, Data Warehousing, ETL, ELT, Query Optimization, Data Normalization
- **Pipeline & Integration** : Data Pipelines, Metadata Driven Pipelines, API Integration, OAuth 2.0, JSON Flattening, Data Transformation, Data Orchestration
- **Languages & Frameworks** : Python, Scala, SQL, Java, PySpark, FastAPI, .NET, Node.js
- **Cloud & DevOps** : Azure, AWS, Docker, Kubernetes, CI CD, Terraform, GitHub Actions, Monitoring, Logging
- **Data Quality & Governance** : Data Security, Data Validation, Data Quality, Access Control, Encryption, Data Governance

WORK EXPERIENCE

Senior Software Engineer

Lyra Health — **Jun 2023 - Dec 2025**

- Designed and implemented scalable **Azure Data Factory** and **Azure Databricks** pipelines processing multi-source healthcare data, improving ingestion efficiency by 45% while ensuring secure integration with **ADLS** and structured **lakehouse architecture**.
- Built advanced **dimensional models** using **star schema** techniques within **Azure Synapse Analytics**, enabling faster reporting queries by 35% and supporting enterprise analytics requirements aligned with clinical decision systems.
- Developed metadata driven **data orchestration pipelines** using **Spark** and **PySpark**, allowing dynamic pipeline configuration and reducing manual maintenance effort by 40% across multiple data domains.
- Integrated external APIs using **OAuth 2.0** authentication and implemented robust **JSON normalization and flattening** logic into relational structures, improving downstream query performance and enabling consistent analytics consumption.
- Optimized complex **SQL transformations** and query execution plans within **Azure SQL Database**, reducing data latency by 30% while maintaining high data quality and reliability across mission critical pipelines.
- Implemented **data quality frameworks** with automated validation, alerting, and monitoring using logging systems, reducing data inconsistencies by 25% and ensuring compliance with healthcare data standards.
- Built secure and scalable **data lakehouse solutions** combining data lakes and warehouses, improving storage efficiency and enabling unified analytics across structured and semi structured datasets.
- Enhanced pipeline resiliency by introducing retry logic, checkpointing, and distributed execution using **Spark**, ensuring fault tolerant processing for high volume healthcare datasets.

- Mentored junior engineers on **Azure data engineering best practices**, including pipeline design, dimensional modeling, and performance tuning, improving team delivery speed and technical consistency.
- Developed real time ingestion patterns for near real time analytics use cases, supporting clinical insights dashboards with reduced processing latency under heavy data loads.
- Created comprehensive documentation and architecture diagrams for **data workflows**, improving onboarding efficiency and ensuring maintainable and repeatable data engineering solutions.
- Worked closely with analytics and BI teams to deliver reliable data models and pipelines, enabling improved reporting accuracy and driving measurable business outcomes across healthcare operations.

Senior Software Engineer

Color — **Apr 2022 - Jun 2023**

- Implemented scalable **Azure data pipelines** using **Data Factory** and **Spark** to process high volume testing and patient datasets, improving ingestion throughput by 40% under peak demand scenarios.
- Designed and optimized **dimensional data models** for reporting systems, enabling efficient analytics queries and reducing dashboard latency by 30% across public health insights platforms.
- Developed API ingestion pipelines with **OAuth 2.0** authentication, transforming nested **JSON data** into structured relational formats using **SQL** and **PySpark** transformations.
- Built **data orchestration frameworks** using metadata driven approaches, improving pipeline flexibility and reducing deployment time for new data sources by 35%.
- Improved **data quality and validation processes** through automated checks and monitoring systems, reducing downstream data issues and increasing trust in analytics outputs.
- Enhanced scalability of **lakehouse architecture** using **ADLS** and **Databricks**, supporting high velocity data streams and enabling near real time reporting capabilities.
- Optimized **SQL queries** and indexing strategies across large datasets, significantly improving performance of analytics workloads and reducing compute costs.
- Implemented secure **data access controls** and governance policies aligned with healthcare compliance standards, ensuring safe handling of sensitive data.
- Contributed to internal training sessions on **Azure data engineering best practices**, helping improve team knowledge and adoption of modern data tools.
- Supported cross functional teams by delivering reliable and well documented **data solutions**, improving data accessibility and enabling better decision making.

Software Engineer 2

Microsoft — **Feb 2020 - Jun 2021**

- Developed scalable **data pipelines** using **Azure Data Factory** and **Azure SQL Database**, enabling ingestion and transformation of enterprise datasets with improved reliability and reduced failure rates.
- Implemented **ETL workflows** using **SQL** and **Python**, optimizing data transformation logic and improving data processing efficiency by 25% across reporting systems.
- Designed initial **dimensional models** supporting reporting use cases, improving query performance and enabling structured analytics for business intelligence teams.
- Integrated third party APIs with secure authentication and transformed semi structured data into normalized relational formats using **JSON processing techniques**.
- Improved pipeline monitoring and logging systems, reducing debugging time and increasing system observability across distributed data workflows.
- Worked on performance tuning of **SQL queries** and database operations, ensuring efficient handling of growing data volumes and improving system responsiveness.
- Contributed to implementation of **data governance and security practices**, ensuring compliance with enterprise standards and protecting sensitive data assets.
- Supported migration of legacy systems to **Azure cloud data solutions**, improving scalability and reducing

infrastructure maintenance overhead.

Software Engineer

Microsoft **Dec 2016 - Feb 2020**

- Built foundational **ETL pipelines** using **SQL** and **Python**, enabling structured data ingestion and transformation for internal reporting systems with improved consistency.
- Developed early stage **data models** supporting business reporting needs, improving data accessibility and simplifying analytics workflows.
- Implemented data ingestion processes from APIs and flat files, transforming raw data into normalized formats suitable for relational databases.
- Optimized database performance through indexing and query tuning, improving response times for reporting applications handling growing datasets.
- Supported transition to cloud based data systems by contributing to migration efforts and improving scalability of data processing workflows.
- Assisted in establishing **data quality checks** and validation processes, ensuring accuracy and reliability of reporting data across internal systems.

EDUCATION

Bachelor's Degree in Computer Science
University of California, Davis **Sep 2012 – Jun 2016**

LICENSES & CERTIFICATIONS

- AWS Certified Developer – Associate — Amazon Web Services (Jun 2022)
- Google Cloud Professional Cloud Architect — Google Cloud (Sep 2021)
- Certified Kubernetes Administrator (CKA) — Cloud Native Computing Foundation (Mar 2021)
- Docker Certified Associate (DCA) — Docker (Jan 2020)